

MATH 4820 / Regression Analysis

Instructor: Mehdi Maadooliat

CHAPTER 2

Simple Linear Regression



Department of Mathematics, Statistics and Computer Science

2.1 SIMPLE LINEAR REGRESSION MODEL

- Single regressor, x ; response, y

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Population
regression model

- β_0 – intercept: if $x = 0$ is in the range, then β_0 is the mean of the response y , when $x = 0$; if $x = 0$ is not in the range, then β_0 has no practical interpretation
- β_1 – slope: change in the mean of the response produced by a unit change in x
- ε – random error

2.1 SIMPLE LINEAR REGRESSION MODEL

- The response, y , is a random variable
- There is a probability distribution for y at each value of x

– Mean:

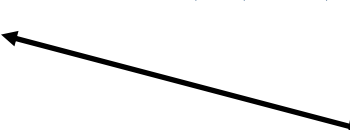
$$E(y|x) = E(\beta_0 + \beta_1 x + \varepsilon) = \beta_0 + \beta_1 x$$

– Variance:

$$Var(y|x) = Var(\beta_0 + \beta_1 x + \varepsilon) = \sigma^2$$

- β_0 and β_1 are unknown and must be estimated

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, 2, \dots, n$$



Sample
regression
model

2.2 LEAST-SQUARES ESTIMATION OF THE PARAMETERS

- Least squares estimation seeks to minimize the *sum of squares* of the differences between the observed response, y_i , and the straight line, $\beta_0 + \beta_1 x_i$.

$$S(\beta_0, \beta_1) = \sum_i \varepsilon_i^2 = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$$

- Let $\hat{\beta}_0, \hat{\beta}_1$ represent the least squares estimators of β_0 and β_1 , respectively.
- These estimators must satisfy:

$$\left. \frac{\partial S}{\partial \beta_0} \right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\left. \frac{\partial S}{\partial \beta_1} \right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0$$

2.2 LEAST-SQUARES ESTIMATION OF THE PARAMETERS

- Simplifying yields the least squares **normal equations**:

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$$

$$\hat{\beta}_0 \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n y_i x_i$$

- Solving the normal equations yields the ordinary least squares estimators:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \qquad \hat{\beta}_1 = \frac{\sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i\right)\left(\sum_{i=1}^n x_i\right)}{n}}{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}}$$

2.2 LEAST-SQUARES ESTIMATION OF THE PARAMETERS

- The fitted simple linear regression model:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

- Sum of Squares Notation:

$$S_{xx} = \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n} = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$S_{xy} = \sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i\right)\left(\sum_{i=1}^n x_i\right)}{n} = \sum_{i=1}^n y_i (x_i - \bar{x})$$

2.2 LEAST-SQUARES ESTIMATION OF THE PARAMETERS

- Then

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i\right)\left(\sum_{i=1}^n x_i\right)}{n}}{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}} = \frac{S_{xy}}{S_{xx}}$$

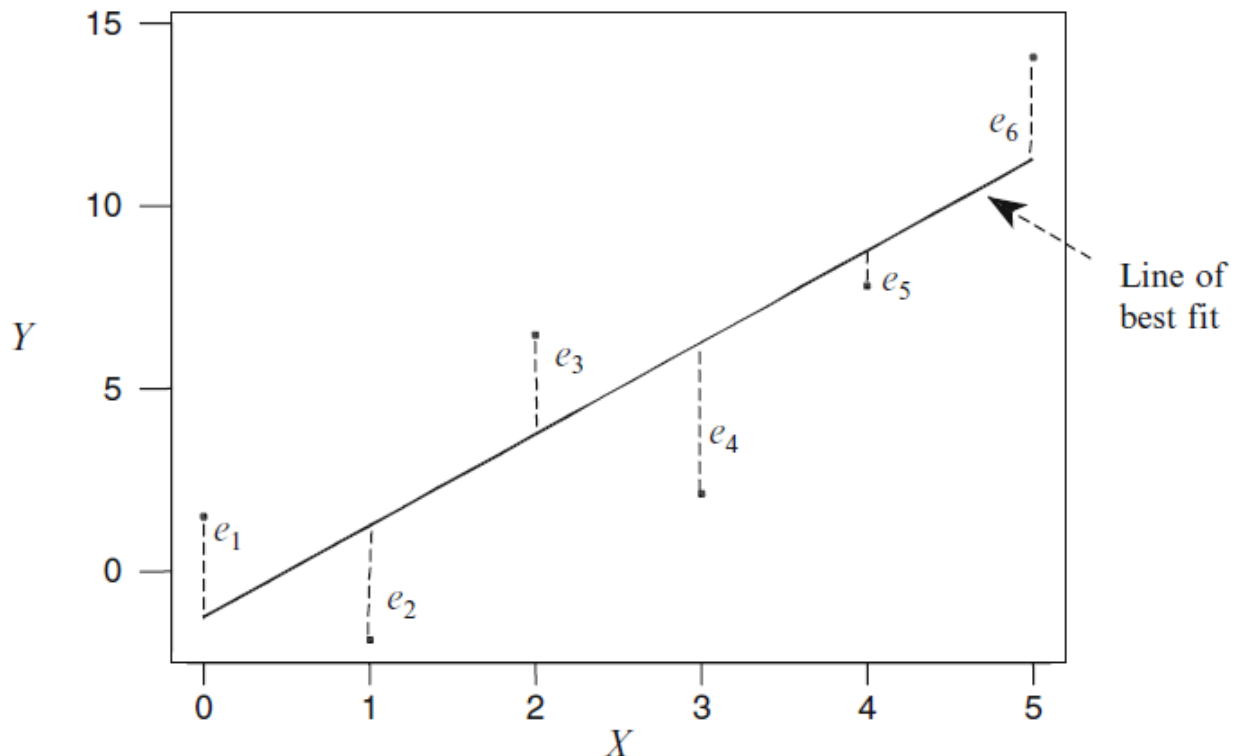
- If $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, for n pairs of (x_i, y_i) s
- Let's define $\bar{x} = 1/n \sum_i x_i$ and $\bar{y} = 1/n \sum_i y_i$
 - $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$
 - $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$
- We can estimate the regression coefficients as:
 - $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$
 - $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

2.2 LEAST-SQUARES ESTIMATION OF THE PARAMETERS

- Residuals:

$$e_i = y_i - \hat{y}_i$$

- Residuals will be used to determine the **adequacy** of the model



MARQUETTE
UNIVERSITY
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EXAMPLE 2.1- ROCKET PROPELLANT DATA

$$S_{xx} = \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n} = 4677.69 - \frac{71,422.56}{20} = 1106.56$$

$$\begin{aligned} S_{xy} &= \sum_{i=1}^n x_i y_i - \frac{\sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n} = 528,492.64 - \frac{(267.25)(42,627.15)}{20} \\ &= -41,112.65 \end{aligned}$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-41,112.65}{1106.56} = -37.15$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 2131.3575 - (-37.15)13.3625 = 2627.82$$

- The fitted least squares regression line is

$$\hat{y} = 2627.82 - 37.15x$$

TABLE 2.2 Data, Fitted Values, and Residuals for Example 2.1

Observed Value, y_i	Fitted Value, $\hat{y}_i$	Residual, e_i
2158.70	2051.94	106.76
1678.15	1745.42	-67.27
2316.00	2330.59	-14.59
2061.30	1996.21	65.09
2207.50	2423.48	-215.98
1708.30	1921.90	-213.60
1784.70	1736.14	48.56
2575.00	2534.94	40.06
2357.90	2349.17	8.73
2256.70	2219.13	37.57
2165.20	2144.83	20.37
2399.55	2488.50	-88.95
1799.80	1698.98	80.82
2336.75	2265.58	71.17
1765.30	1810.44	-45.14
2053.50	1959.06	94.44
2414.40	2404.90	9.50
2200.50	2163.40	37.10
2654.20	2553.52	100.68
1753.70	1829.02	-75.32
$\Sigma y_i = 42627.15$	$\Sigma \hat{y}_i = 42627.15$	$\Sigma e_i = 0.00$

R CODE

- *The R code to get the scatterplot:*
 - `rocket=read.table("data-ex-2-1_(Rocket_Prop).txt", header=T)`
 - `x=rocket$Age`
 - `y=rocket$Strength`
 - `plot(x,y,xlab="Age of propellant, x",ylab="Shear strength, y")`
- *The R code to get the least square estimators:*
 - `rocket.lm=lm(y~x)`
 - `summary(rocket.lm)`
 - `rocket.lm$fitted.values`
- *(Note: lm() is the linear model function)*

2.2 LEAST-SQUARES ESTIMATION OF THE PARAMETERS

R output

```
Call:
lm(formula = y ~ x)

Residuals:
    Min       1Q   Median       3Q      Max
-215.98  -50.68   28.74   66.61  106.76

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2627.822     44.184   59.48  < 2e-16 ***
x            -37.154       2.889  -12.86 1.64e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 96.11 on 18 degrees of freedom
Multiple R-squared:  0.9018,    Adjusted R-squared:  0.8964
F-statistic: 165.4 on 1 and 18 DF,  p-value: 1.643e-10
```

TABLE 2.3 MINITAB Regression Output for Example 2.1

Regression Analysis

The regression equation is
Strength = 2628 - 37.2 Age

Predictor	Coef	StDev	T	P
Constant	2627.82	44.18	59.47	0.000
Age	-37.154	2.889	-12.86	0.000

S = 96.11 R-Sq = 90.2% R-Sq(adj) = 89.6%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	1527483	1527483	165.38	0.000
Error	18	166255	9236		
Total	19	1693738			

Minitab

2.2.2 PROPERTIES OF THE LEAST-SQUARES ESTIMATORS AND THE FITTED REGRESSION MODEL

- The ordinary least-squares (OLS) estimator of the slope is a linear combinations of the observations, y_i :

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \sum_{i=1}^n c_i y_i$$

where

- $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$
- $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n (x_i - \bar{x})y_i$

Therefore

- $c_i = \frac{(x_i - \bar{x})}{S_{xx}}$
 - $\sum_{i=1}^n c_i = 0$
 - $\sum_{i=1}^n c_i x_i = 1$
- Useful in calculating expected value and variance properties

2.2.2 PROPERTIES OF THE LEAST-SQUARES ESTIMATORS AND THE FITTED REGRESSION MODEL

- The least-squares estimators are **unbiased estimators** of their respective parameter:

$$E(\hat{\beta}_1) = \beta_1 \quad E(\hat{\beta}_0) = \beta_0$$

- The variances are

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}} \quad Var(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$$

- The OLS estimators are **Best Linear Unbiased Estimators (BLUE)**
- “Best” implies minimum variance among unbiased estimators under the certain assumptions.

2.2.2 PROPERTIES OF THE LEAST-SQUARES ESTIMATORS AND THE FITTED REGRESSION MODEL

Useful properties of the least-squares fit

1. $\sum_{i=1}^n (y_i - \hat{y}_i) = \sum_{i=1}^n e_i = 0$
2. $\sum_{i=1}^n y_i = \sum_{i=1}^n \hat{y}_i$
3. The least-squares regression line always passes through the centroid $(\bar{x}, \bar{y})$ of the data.
4. $\sum_{i=1}^n x_i e_i = 0$
5. $\sum_{i=1}^n \hat{y}_i e_i = 0$

2.2.3 ESTIMATION OF σ^2

- Residual (error) sum of squares

$$SS_{Res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2$$

- Unbiased estimator of σ^2 : (see Section C.3)

$$\hat{\sigma}^2 = \frac{SS_{Res}}{n - 2} = MS_{Res}$$

- The quantity MS_{Res} is called the **residual mean square**.
- The square root of MS_{Res} is sometimes called the **standard error** of regression.
- The quantity $n - 2$ is the number of **degrees of freedom** for the SS_{Res}

- SS_{Res} can be further simplified for convenient computing :

$$\begin{aligned} SS_{Res} &= \underbrace{\sum_{i=1}^n y_i^2 - n\bar{y}^2}_{SS_T = \sum (y_i - \bar{y})^2} - \hat{\beta}_1 S_{xy} \\ &= SS_T - \hat{\beta}_1 S_{xy} \end{aligned}$$

2.2.3 ESTIMATION OF σ^2

Estimates of the $\hat{\sigma}^2$ depends on the residual sum of squares.

$$\hat{\sigma}^2 = \frac{SS_{Res}}{n - 2} = MS_{Res}$$

Then:

- Any violation of the assumptions on the model errors could damage the usefulness of this estimate
- A misspecification of the model can damage the usefulness of this estimate
- This estimate is **model dependent**

2.3 HYPOTHESIS TESTING ON THE SLOPE AND INTERCEPT

- Three assumptions needed to apply procedures such as **hypothesis testing** and **confidence intervals**.

Model errors, ε_i ,

- are **normally distributed**
- are **independently** distributed
- have **constant variance**

i.e. $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$. This can be used to show

- $Z = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2 / S_{xx}}} \sim N(0, 1)$, and
- $t = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{MS_{Res} / S_{xx}}} \sim t_{n-2}$

which can be used for hypothesis testing and confidence intervals.

2.3.1 USE OF T-TESTS

- T-test for Slope:

$$H_0: \beta_1 = \beta_{10} \quad H_1: \beta_1 \neq \beta_{10}$$

- Standard error of the slope: $se(\hat{\beta}_1) = \sqrt{\frac{MS_{Res}}{S_{xx}}}$
- Test statistic: $t_0 = \frac{\hat{\beta}_1 - \beta_{10}}{se(\hat{\beta}_1)}$
- Reject H_0 if $|t_0| > t_{\alpha/2, n-2}$
- Can also use the P -value approach:
 $p\text{-value} = 2P(t \geq |t_0|)$

2.3.1 USE OF T-TESTS

- T-test for Intercept:

$$H_0: \beta_0 = \beta_{00} \quad H_1: \beta_0 \neq \beta_{00}$$

- Standard error of the intercept: $se(\hat{\beta}_0) = \sqrt{MS_{\text{Res}} \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)}$

- Test statistic: $t_0 = \frac{\hat{\beta}_0 - \beta_{00}}{se(\hat{\beta}_0)}$

- Reject H_0 if $|t_0| > t_{\alpha/2, n-2}$

- Can also use the P -value approach:

$$\text{p-value} = 2P(t \geq |t_0|)$$

2.3.2 TESTING SIGNIFICANCE OF REGRESSION

$$H_0: \beta_1 = 0 \quad \text{vs} \quad H_1: \beta_1 \neq 0$$

- This tests the **significance of regression**; that is, is there a linear relationship between the response and the regressor?
- *Failing to reject $\beta_1 = 0$* , implies that there is no linear relationship between y and x

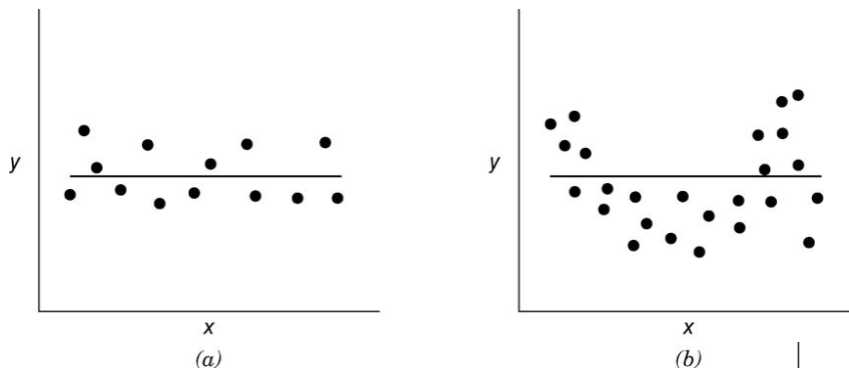


Figure 2.2 Situations where the hypothesis $H_0: \beta_1 = 0$ is not rejected.

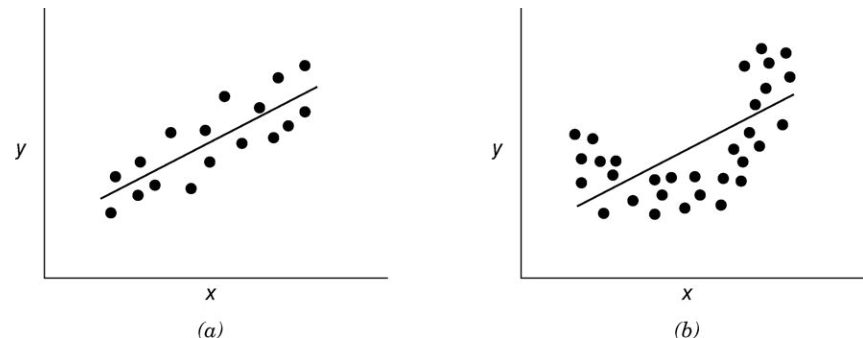


Figure 2.3 Situations where the hypothesis $H_0: \beta_1 = 0$ is rejected.

EXAMPLE 2.3- ROCKET PROPELLANT DATA

Example 2.3 The Rocket Propellant Data

We test for significance of regression in the rocket propellant regression model of [Example 2.1](#). The estimate of the slope is $\hat{\beta}_1 = -37.15$, and in [Example 2.2](#), we computed the estimate of σ^2 to be $MS_{\text{Res}} = \hat{\sigma}^2 = 9244.59$. The standard error of the slope is

$$\text{se}(\hat{\beta}_1) = \sqrt{\frac{MS_{\text{Res}}}{S_{xx}}} = \sqrt{\frac{9244.59}{1106.56}} = 2.89$$

Therefore, the test statistic is

$$t_0 = \frac{\hat{\beta}_1}{\text{se}(\hat{\beta}_1)} = \frac{-37.15}{2.89} = -12.85$$

If we choose $\alpha = 0.05$, the critical value of t is $t_{0.025,18} = 2.101$. Thus, we would reject $H_0: \beta_1 = 0$ and conclude that there is a linear relationship between shear strength and the age of the propellant.

2.3.3 THE ANALYSIS OF VARIANCE APPROACH AN ALTERNATIVE APPROACH FOR HYP. TEST

- Partitioning of total variability

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i)$$

$$\begin{aligned}\sum (y_i - \bar{y})^2 &= \sum (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y}_i)^2 \\ &\quad + 2\sum (\hat{y}_i - \bar{y})(y_i - \hat{y}_i)\end{aligned}$$

or

$$\underbrace{\sum (y_i - \bar{y})^2}_{SS_T} = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SS_R} + \underbrace{\sum (y_i - \hat{y}_i)^2}_{SS_{Res}}$$

- It can be shown that $SS_R = \hat{\beta}_1 S_{xy}$
- Degrees of Freedom:

$$\underbrace{n-1}_{df_T} = \underbrace{1}_{df_R} + \underbrace{(n-2)}_{df_{Res}}$$

2.3.3 THE ANALYSIS OF VARIANCE

- Mean Squares:

$$MS_R = \frac{SS_R}{1} \quad MS_{Res} = \frac{SS_{Res}}{n-2}$$

- ANOVA procedure for testing $H_0: \beta_1 = 0$

Source of Variation	Sum of Squares	DF	MS	F_0
Regression	SS_R	1	MS_R	MS_R/MS_{Res}
Residual	SS_{Res}	$n-2$	MS_{Res}	
Total	SS_T	$n-1$		

- A large value of F_0 indicates that regression is significant; specifically, reject H_0 if $F_0 > F_{\alpha,1,n-2}$
- Can also use the P -value approach:

$$p\text{-value} = P(F \geq F_0)$$

R CODE

➤ `rocket=read.table("(Rocket_Prop).txt", header=T)`

➤ `y=rocket$Strength`

➤ `x=rocket$Age`

➤ `rocket.lm=lm(y~x)`

➤ `anova(rocket.lm)`

Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
x	1	1527483	1527483	165.38	1.643e-10 ***
Residuals	18	166255	9236		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

TABLE 2.5 Analysis-of-Variance Table for the Rocket Propellant Regression Model

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P value
Regression	1,527,334.95	1	1,527,334.95	165.21	1.66×10^{-10}
Residual	166,402.65	18	9,244.59		
Total	1,693,737.60	19			

2.3.3 THE ANALYSIS OF VARIANCE

Relationship between t_0 and F_0 :

- For $H_0: \beta_1 = 0$, it can be shown that

$$t_0 = \frac{\hat{\beta}_1}{\text{se}(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{\sqrt{MS_{Res}/S_{xx}}}$$

- Using the fact $SS_R = \hat{\beta}_1 S_{xy}$, we have:

$$t_0^2 = \frac{\hat{\beta}_1^2 S_{xx}}{MS_{Res}} = \frac{\hat{\beta}_1 S_{xy}}{MS_{Res}} = \frac{MS_R}{MS_{Res}}$$

- This means that $t_0^2 = F_0$.
- So for testing significance of regression, the t-test and the ANOVA procedure are equivalent
(*only true in simple linear regression*)

2.4 INTERVAL ESTIMATION IN SIMPLE LINEAR REGRESSION

- 100(1- α)% Confidence interval for Slope

$$\hat{\beta}_1 - t_{\alpha/2, n-2} se(\hat{\beta}_1) \leq \beta_1 \leq \hat{\beta}_1 + t_{\alpha/2, n-2} se(\hat{\beta}_1)$$

- 100(1- α)% Confidence interval for the Intercept

$$\hat{\beta}_0 - t_{\alpha/2, n-2} se(\hat{\beta}_0) \leq \beta_0 \leq \hat{\beta}_0 + t_{\alpha/2, n-2} se(\hat{\beta}_0)$$

Example 2.5 The Rocket Propellant Data

We construct 95% CIs on β_1 and σ^2 using the rocket propellant data from Example 2.1. The standard error of $\hat{\beta}_1$ is $se(\hat{\beta}_1) = 2.89$ and $t_{0.025, 18} = 2.101$. Therefore, from Eq. (2.35), the 95% CI on the slope is

$$\begin{aligned} \hat{\beta}_1 - t_{0.025, 18} se(\hat{\beta}_1) &\leq \beta_1 \leq \hat{\beta}_1 + t_{0.025, 18} se(\hat{\beta}_1) \\ -37.15 - (2.101)(2.89) &\leq \beta_1 \leq -37.15 + (2.101)(2.89) \end{aligned}$$

or

$$-43.22 \leq \beta_1 \leq -31.08$$

2.4 INTERVAL ESTIMATION IN SIMPLE LINEAR REGRESSION

- $100(1 - \alpha)\%$ Confidence interval for σ^2 :

$$\frac{(n - 2)MS_{RES}}{\chi_{\alpha/2, n-2}^2} \leq \sigma^2 \leq \frac{(n - 2)MS_{RES}}{\chi_{1-\alpha/2, n-2}^2}$$

- If the errors are normally and independently distributed, Appendix C.3 shows that the sampling distribution of

$$\frac{(n - 2)MS_{Res}}{\sigma^2}$$

is following chi-squared with $n - 2$ degrees of freedom.

2.4 INTERVAL ESTIMATION IN SIMPLE LINEAR REGRESSION

- Let x_0 be the level of the regressor variable at which we want to estimate the mean response, i.e.

$$E(y|x_0) = \mu_{y|x_0}$$

- Point estimator of $E(y|x_0)$ once the model is fit:

$$E(\widehat{y|x_0}) = \hat{\mu}_{y|x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

- In order to construct a confidence interval on the mean response, we need the variance of the point estimator.

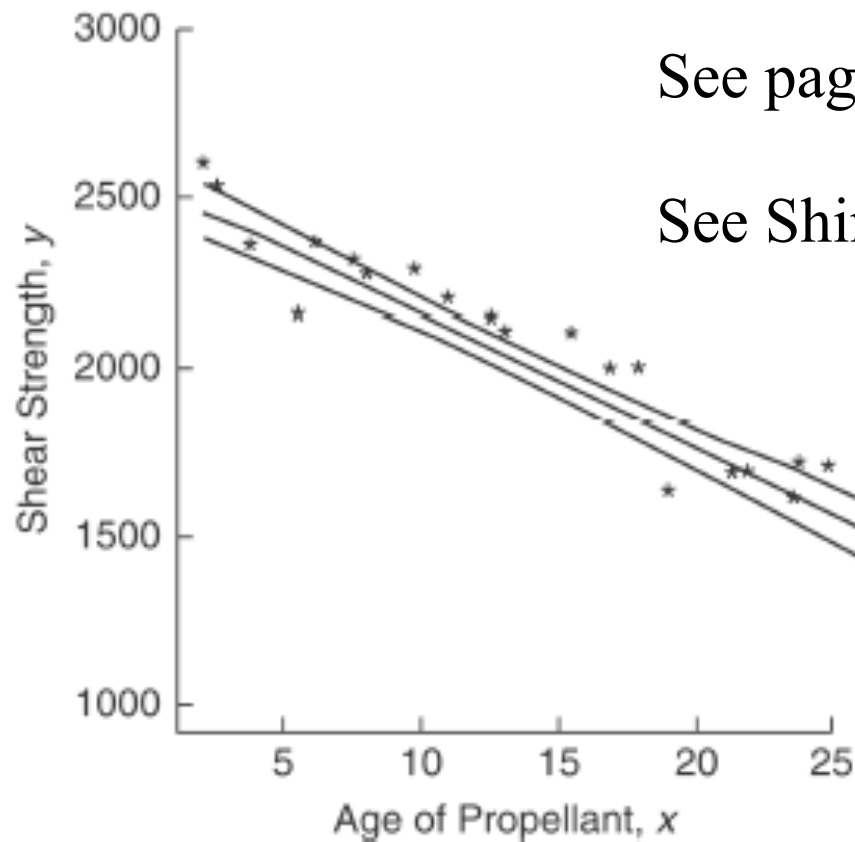
$$\begin{aligned} Var(\hat{\mu}_{y|x_0}) &= Var(\hat{\beta}_0 + \hat{\beta}_1 x_0) = Var[\bar{y} + \hat{\beta}_1(x_0 - \bar{x})] \\ &= Var(\bar{y}) + Var[\hat{\beta}_1(x_0 - \bar{x})] \\ &= \frac{\sigma^2}{n} + \frac{\sigma^2(x_0 - \bar{x})^2}{S_{xx}} = \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right] \end{aligned}$$

2.4.2 INTERVAL ESTIMATION OF THE MEAN RESPONSE

- 100(1- α)% confidence interval for $E(y|x_0)$

$$\begin{aligned}\hat{\mu}_{y|x_0} - t_{\alpha/2, n-2} \sqrt{MS_{\text{Res}} \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]} &\leq E(y | x_0) \\ &\leq \hat{\mu}_{y|x_0} + t_{\alpha/2, n-2} \sqrt{MS_{\text{Res}} \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]}\end{aligned}$$

Notice that the **length** of the CI depends on the location of the point of interest



See pages 34-35, text

See Shiny app

Figure 2.4 The upper and lower 95% confidence limits for the propellant data.

2.5 PREDICTION OF NEW OBSERVATIONS

- Suppose we wish to construct a *prediction* interval on a future observation, y_0 corresponding to a particular level of x , say x_0 .
- The point estimate would be:

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

- The confidence interval on the mean response at this point is not appropriate for this situation. Why?

2.5 PREDICTION OF NEW OBSERVATIONS

- Let the random variable, Ψ be $\Psi = y_0 - \hat{y}_0$
- Ψ is normally distributed with

- $E(\Psi) = 0$
 - $Var(\Psi) = Var(y_0 - \hat{y}_0) = \sigma^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]$

(y_0 is independent of $\hat{y}_0$)

- 100(1 - α)% prediction interval on a future observation, y_0 , at x_0

$$\hat{y}_0 - t_{\alpha/2, n-2} \sqrt{MS_{\text{Res}} \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]} \leq y_0$$

$$\leq \hat{y}_0 + t_{\alpha/2, n-2} \sqrt{MS_{\text{Res}} \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]}$$

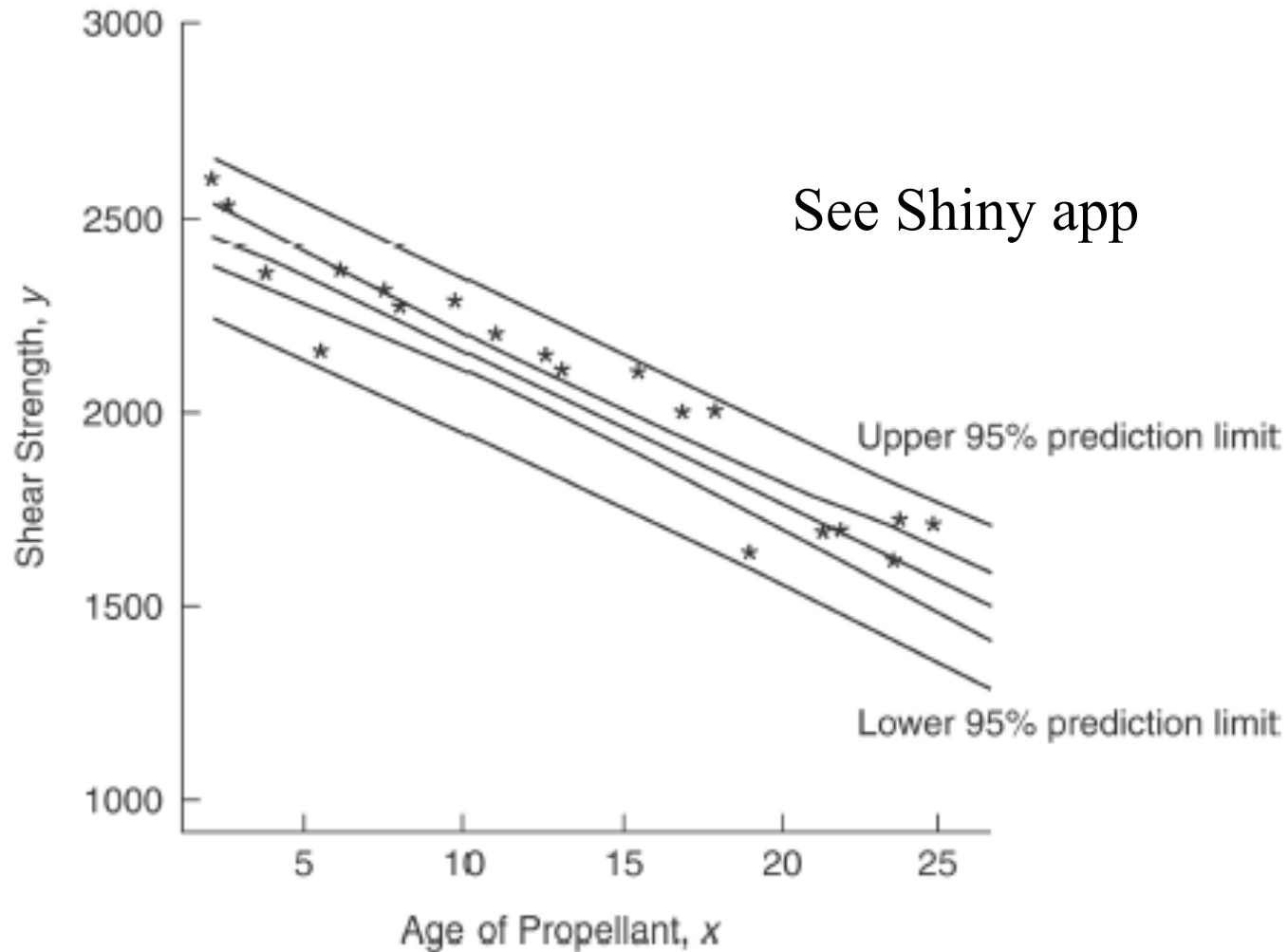


Figure 2.5 The 95% confidence and prediction intervals for the propellant data.

R CODE

- ##### confidence interval on beta0 and beta1 #####
 - `rocket.lm <- lm(y~x, data=rocket.prop)`
 - `confint(rocket.lm,level=0.95)`
- ##### CI on the mean response at the point $x=x_0$ #####
- ##### Example 2.6
 - `newdata=data.frame(x=13.3625)`
 - `predict(rocket.lm, newdata,interval="confidence")`
 - `conf.i <- predict(rocket.lm, interval=c("confidence"))`
- ##### prediction of new observation, prediction interval #####
- ##### Example 2.7
 - `newdata=data.frame(x=10)`
 - `predict(rocket.lm, newdata, interval="prediction")`
 - `newx = seq(0,30,by = 0.1); plot(rocket.prop[,2:1])`
 - `conf_interval <- predict(rocket.lm, newdata=data.frame(x=newx), interval="prediction", level = 0.95)`
 - `matlines(newx, conf_interval[,2:3], col = "blue", lty=2)`

2.6 COEFFICIENT OF DETERMINATION

- $R^2 = \frac{SS_R}{SS_T}$ - coefficient of determination
- Since
 - SS_T is a measure of the variability in y without considering the effect of the regressor variable x and
 - SS_{Res} is a measure of the variability in y remaining after x has been considered
- $R^2 = 1 - \frac{SS_{Res}}{SS_T}$ is often called the *proportion of variation* explained by the regressor x
- For the rocket propellant data

$$R^2 = \frac{SS_R}{SS_T} = \frac{1,527,334.95}{1,693,737.60} = 0.9018$$

2.6 COEFFICIENT OF DETERMINATION

- R^2 can be misleading!
 - Simply adding more terms to the model will increase R^2
 - As the range of the regressor variable increases (decreases), R^2 generally increases (decreases).
 - R^2 does not indicate the appropriateness of a linear model

- R code:

➤ `summary(rocket.lm)`

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 2627.822      44.184   59.48 < 2e-16 ***
x            -37.154       2.889  -12.86 1.64e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 96.11 on 18 degrees of freedom
Multiple R-squared:  0.9018,    Adjusted R-squared:  0.8964
F-statistic: 165.4 on 1 and 18 DF,  p-value: 1.643e-10
```

2.9 CONSIDERATIONS IN THE USE OF REGRESSION

- Extrapolating
- Extreme points will often influence the slope.
- Outliers can disturb the least-squares fit

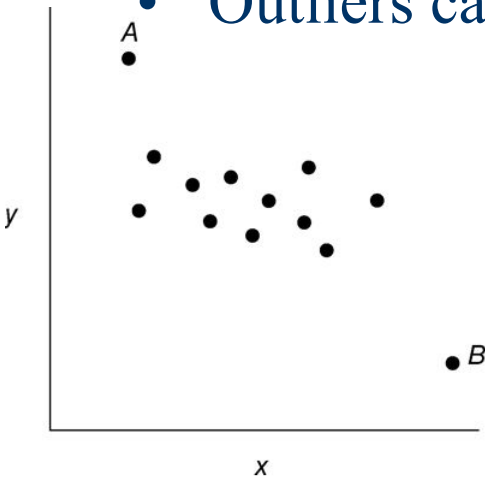


Figure 2.8 Two influential observations.

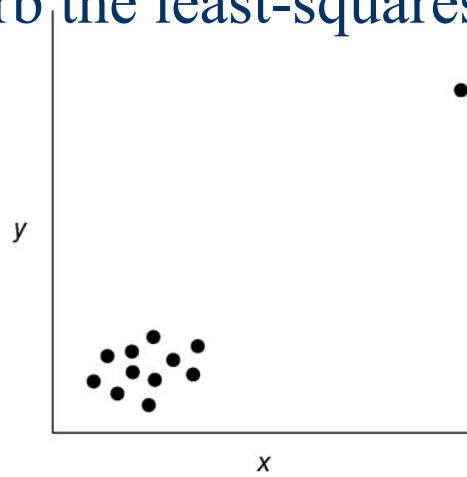


Figure 2.9 A point remote in x space.

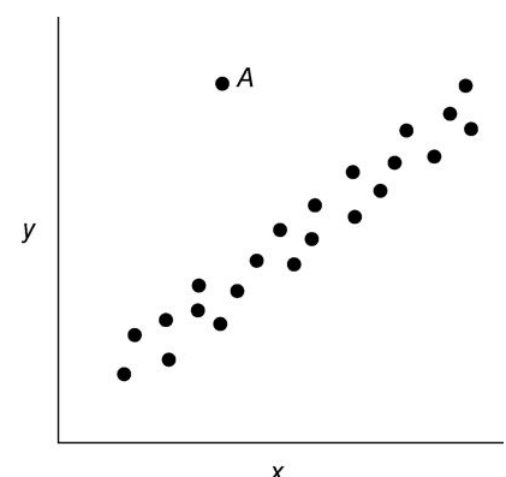


Figure 2.10 An outlier.

- Linear relationship does not imply cause-effect relationship ([spurious correlations](#))
- Sometimes, the value of the regressor variable is unknown and itself be estimated or predicted.

2.11 ESTIMATION BY MAXIMUM LIKELIHOOD

- The method of least squares is not the only method of parameter estimation that can be used for linear regression models.
- If the form of the response (error) distribution is known, then the **maximum likelihood (ML)** estimation method can be used.
- In simple linear regression with normal errors, the MLEs for the regression coefficients are identical to that obtained by the method of least squares.
- Details of finding the MLEs are on page 51 of the book.
- The MLE of σ^2 is *biased*. (Not a serious problem here though.)

Consider the data $(y_i, x_i), i = 1, 2, \dots, n$. If we assume that the errors in the regression model are $NID(0, \sigma^2)$, then the observations y_i in this sample are normally and independently distributed random variables with mean $\beta_0 + \beta_1 x_i$ and variance σ^2 . The likelihood function is found from the joint distribution of the observations. If we consider this joint distribution with the observations given and the parameters β_0, β_1 , and σ^2 unknown constants, we have the likelihood function. For the simple linear regression model with normal errors, the **likelihood function** is

$$\begin{aligned} L(y_i, x_i, \beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^n (2\pi\sigma^2)^{-1/2} \exp\left[-\frac{1}{2\sigma^2} (y_i - \beta_0 - \beta_1 x_i)^2\right] \\ &= (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right] \end{aligned} \quad (2.56)$$

The maximum-likelihood estimators are the parameter values, say $\tilde{\beta}_0, \tilde{\beta}_1$, and $\tilde{\sigma}^2$, that maximize L , or equivalently, $\ln L$. Thus,

$$\begin{aligned} \ln L(y_i, x_i, \beta_0, \beta_1, \sigma^2) &= -\left(\frac{n}{2}\right) \ln 2\pi - \left(\frac{n}{2}\right) \ln \sigma^2 \\ &\quad - \left(\frac{1}{2\sigma^2}\right) \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \end{aligned} \quad (2.57)$$

maximum-likelihood estimators $\tilde{\beta}_0$, $\tilde{\beta}_1$, and $\tilde{\sigma}^2$ must satisfy

$$\left. \frac{\partial \ln L}{\partial \beta_0} \right|_{\tilde{\beta}_0, \tilde{\beta}_1, \tilde{\sigma}^2} = \frac{1}{\tilde{\sigma}^2} \sum_{i=1}^n (y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i) = 0 \quad (2.58a)$$

$$\left. \frac{\partial \ln L}{\partial \beta_1} \right|_{\tilde{\beta}_0, \tilde{\beta}_1, \tilde{\sigma}^2} = \frac{1}{\tilde{\sigma}^2} \sum_{i=1}^n (y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i) x_i = 0 \quad (2.58b)$$

and

$$\left. \frac{\partial \ln L}{\partial \sigma^2} \right|_{\tilde{\beta}_0, \tilde{\beta}_1, \tilde{\sigma}^2} = -\frac{n}{2\tilde{\sigma}^2} + \frac{1}{2\tilde{\sigma}^4} \sum_{i=1}^n (y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i)^2 = 0 \quad (2.58c)$$

The solution to [Eq. \(2.58\)](#) gives the **maximum-likelihood estimators**:

Identical to OLS $\nearrow \tilde{\beta}_0 = \bar{y} - \tilde{\beta}_1 \bar{x} \quad (2.59a)$

$\searrow \tilde{\beta} = \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (2.59b)$

Biased $\nearrow$

$\tilde{\sigma}^2 = \frac{\sum_{i=1}^n (y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i)^2}{n} \quad (2.59c)$

2.11 ESTIMATION BY MAXIMUM LIKELIHOOD

In general, maximum-likelihood estimators have better **statistical properties** than least-squares estimators. The maximum-likelihood estimators are **unbiased** (including $\tilde{\sigma}^2$, which is **asymptotically unbiased**, or unbiased as n becomes large) and have **minimum variance** when compared to **all** other unbiased estimators. They are also **consistent estimators** (consistency is a large-sample property indicating that the estimators differ from the true parameter value by a very small amount as n becomes large), and they are a set of **sufficient statistics** (this implies that the estimators contain all of the “information” in the original sample of size n). On the other hand, maximum-likelihood estimation requires more stringent statistical assumptions than the least-squares estimators. The least-squares estimators require only second-moment assumptions (assumptions about the expected value, the variances, and the covariances among the random errors). The maximum-likelihood estimators require a full distributional assumption, in this case that the random errors follow a normal distribution with the same second moments as required for the least-squares estimates. For more information on maximum-likelihood estimation in regression models, see Graybill [1961, 1976], Myers [1990], Searle [1971], and Seber [1977].

2.12 CASE WHERE THE REGRESSOR IS RANDOM

- Suppose that x and y are jointly distributed random variables but the form of this joint distribution is unknown.
- It can be shown that all of our previous regression results hold if the following conditions are satisfied:
 1. The conditional distribution of y given x is normal with conditional mean $\beta_0 + \beta_1 x$ and conditional variance σ^2 .
 2. The x 's are independent random variables whose probability distribution does not involve β_0 , β_1 , and σ^2 .

QUESTIONS?

- ANY QUESTION?